

Title

Reinforcement Learning–Based Anomaly Detection in Context-Aware Heart Rate Monitoring Systems

Abstract

Physiological anomaly detection is an important application in the early diagnosis of health risks and unusual bodily conditions. Conventional machine learning methods tend to use centralized processing and static models, which might not be adaptable to individual-specific physiological variations and could pose privacy and latency issues. This paper proposes an adaptive on-device DQN framework for physiological anomaly detection, with a focus on a new reward adjustment mechanism that can adaptively adjust to varying physiological patterns. Instead of making a new algorithmic contribution, the novelty of the proposed method is in combining on-device learning constraints with an adaptive reward mechanism that can enhance anomaly detection sensitivity and personalization. The proposed method is tested with physiological signal data under simulated device conditions and shows its feasibility and efficacy. The experimental results show that adaptive reward adjustment can improve detection performance over static reward mechanisms. Physical deployment on wearable hardware is identified as future work.

Keywords: DQN, Physiological Signals, Anomaly Detection, On-device Learning, Adaptive Reward Mechanism

1. Introduction

The availability of physiological data through wearable and personal monitoring systems has made it possible to continuously monitor health and detect anomalies early. Heart rate and other physiological signals are useful for gaining insights into the physiological state of an individual. However, it is difficult to detect anomalies in these signals because of the high variability among individuals, noise, and non-stationarity. Most of the existing methods use supervised or unsupervised learning models that are trained in centralized

settings. Although these methods are efficient, they are not adaptable to individual-level changes and pose privacy issues when personal health data is sent to cloud servers. Furthermore, traditional models are not capable of handling dynamic physiological changes due to stress, activity, or health changes. Reinforcement Learning (RL) is a promising technique that allows systems to learn from interactions and feedback received from the environment. RL is particularly suitable for dealing with temporal dependencies as well as adapting decision-making policies over time. In recent years, on-device learning has emerged as a promising solution for dealing with challenges associated with latency, data privacy, and adaptation. This paper describes an adaptive on-device reinforcement learning framework for physiological anomaly detection using Deep Q-Learning (DQN). The main contribution of this paper does not lie in the development of a new algorithm but in the effective combination of on-device reinforcement learning with a reward function dynamically adjusted according to the characteristics of physiological signals. The proposed adaptive approach allows the system to adapt to changing signal patterns while being subject to practical device-level computational limitations.

2. Related Work

Reinforcement learning (RL) has been applied in several healthcare-oriented studies to enable adaptive and personalized decision-making. Yom-Tov et al. employed Q-learning to encourage physical activity among diabetes patients through personalized goal settings, demonstrating the effectiveness of adaptive feedback in influencing user behavior. Other research has utilized Q-learning in Wireless Body Area Networks (WBANs) to improve energy efficiency by optimizing sensor data transmission, focusing primarily on system-level performance rather than health data interpretation. Reinforcement learning has also been explored for time-series anomaly detection, where RL-based and active learning methods adaptively identify anomalous patterns in sequential data. Additionally, recent studies have proposed reinforcement learning-based dynamic health monitoring frameworks that adjust monitoring strategies based on user activity; however, such approaches often rely on complex models and non-physiological inputs, such as video data, which limit their applicability in resource-constrained environments. Overall, existing works either emphasize behavioral intervention, communication efficiency, or computationally intensive architectures, leaving a gap in lightweight, on-device reinforcement learning approaches specifically designed for physiological anomaly detection.

3. System Overview

The proposed framework is intended for on-device implementation and functions under limited computational and memory capabilities. The system analyzes physiological signals on the device and uses a DQN-based reinforcement learning agent to detect anomalies.

3.1 Architecture

The system consists of the following components:

- Physiological signal input module
- Feature extraction and normalization
- DQN agent
- Adaptive reward-adjustment mechanism
- Rule-based validation layer
- Anomaly decision module

All processing is performed locally without reliance on cloud-based inference.

3.2 Rule-Based Validation Layer

In addition to the DQN-based learning agent, a lightweight rule-based validation layer is incorporated to improve decision stability and reduce false alarms. This layer applies simple physiological constraints, such as plausible heart rate ranges and short-term consistency checks, to refine the agent's output. The rule-based layer does not replace the reinforcement learning component; instead, it acts as a post-decision filter that enhances robustness and interpretability, which is particularly important in health monitoring applications.

4. Reinforcement Learning Formulation

The anomaly detection task is modeled as a reinforcement learning problem defined by the tuple (S, A, R, π) .

State (S):

The state represents a feature vector derived from physiological signals, including statistical and temporal characteristics.

Action (A):

The agent selects actions corresponding to anomaly or normal classifications, or confidence-adjusted decisions.

Reward (R):

The reward reflects detection of correctness and adaptation to physiological variations.

5. Adaptive Reward-Adjustment Mechanism (Core Contribution)

Unlike static reward functions, the proposed framework employs an adaptive reward-adjustment mechanism. The reward is dynamically modified based on:

- Signal stability
- Recent detection confidence
- Frequency of anomalies

This mechanism allows the agent to:

- Reduce false positives during stable physiological periods
- Increase sensitivity during abrupt signal changes
- Adapt to individual-specific baseline shifts

The novelty of the work lies in this adaptive integration rather than in a standalone algorithmic innovation.

6. On-Device Learning Considerations

To emulate on-device deployment, the system is evaluated under constrained conditions:

- Limited model size
- Restricted memory usage
- Local inference and updates

Training and evaluation are conducted using a simulated embedded environment, validating feasibility without requiring physical hardware deployment. This approach is consistent with early-stage system validation in academic research.

7. Experimental Evaluation

7.1 Dataset

Publicly available physiological datasets are used for evaluation.

7.2 Results

Results demonstrate improved anomaly detection of accuracy and stability when using the adaptive reward mechanism and a Rule Based Layer. Learning curves indicate faster convergence and reduce false alarms compared to static reward approaches.

8. Discussion

The experimental results highlight the effectiveness of integrating adaptive reward mechanisms with on-device reinforcement learning. While the absence of proper physical hardware deployment is acknowledged, the simulated evaluation sufficiently demonstrates feasibility and conceptual validity.

The framework emphasizes personalization and adaptability, which are critical for physiological monitoring systems. The results support the claim that novelty arises from integration and adaptive design rather than isolated components.

9. Limitations and Future Work

The current study does not include physical deployment on wearable hardware. Future work will focus on:

- Real-time deployment on embedded devices
- Long-term user-specific adaptation studies

10. Conclusion

This paper presented an adaptive on-device reinforcement learning framework for physiological anomaly detection. By integrating an adaptive reward-adjustment mechanism with on-device learning constraints and a rule-based layer, the proposed approach addresses key challenges in personalization, privacy, and adaptability. Experimental evaluation demonstrates the potential of the framework, establishing a foundation for future hardware-based deployment, and extended clinical validation.